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MGU-B.VOC (HONOURS) REGULAR EXAMINATIONS,APRIL 2026

SECOND SEMESTER

SDC-MG2SDCBAT100-PROTECTED CULTIVATION OF HORTICULTURAL CROPS

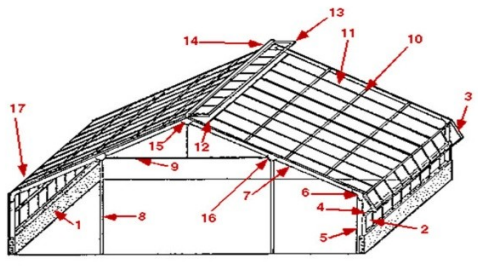
PART A

1. Fogger irrigation is a specialized micro-irrigation technique that uses high-pressure, atomizing nozzles to turn water into an ultra-fine mist (droplets under 0.5mm) to increase humidity, cool down greenhouses, and gently water plants.
2. A rain shelter is a specialized, open-sided structure, often resembling a greenhouse, designed to protect crops from heavy rainfall while allowing proper ventilation and light.
3. Plastic,clay,Terracota,Glass,wood,metal,alternative containers…
4. Fertigation is the technique of injecting fertilizers, soil amendments, and other water-soluble products directly into an irrigation system (primarily drip or sprinkler) to deliver nutrients precisely to the root zone.
5. Harvesting is the agricultural process of gathering mature crops, fruits, vegetables, or plants from the fields once they have reached full growth. It involves cutting, digging, or picking the, edible portions, marking the end of the growing season, and is often the most labor-intensive step in farming.
6. A **North-South (N-S)** orientation generally provides the most uniform sunlight distribution throughout the day.
7. Even span type ,Uneven span type .

PART B

8. Special horticultural practices in cucumber include trellising for vertical growth, pruning (removing lower side shoots/leaves), and using specific chemical sprays to influence flower sex, ensuring high yield and quality, especially in greenhouse systems.
9. Cooling system-fan and pad ,fog cooling system,horizontal air flow .
   Heating system-unit heaters,centralized,infrared,solar
   Ventilation system,microprocessors

10.



1. sidewall or curtain wall
2. lower side sash
3. side ventilator
4. transom sill
5. side post
6. splice plate or gusset
7. rafter
8. column
9. cross tie
10. purlin
11. sash bar
12. ridge vent header
13. ridge vent
14. ridge and ridge cap
15. ridge plate
16. column and tie plate
17. eave

11.A **unit heater** is a self-contained, non-ducted device that heats the specific area where it is installed. It uses a fuel source (like natural gas, propane, or electricity) to heat a heat exchanger. A powerful internal fan then blows air over that exchanger and directly into the room.

Boilers are the heart of a greenhouse's **central heating system**, typically chosen for larger commercial operations (over 1,000 m^2) or facilities requiring precise temperature control. Unlike unit heaters that blow hot air, boilers use water or steam to transport heat, which is up to **3,500 times more efficient** at carrying energy than air.

12. **Convection Tube Cooling:** A fan introduces cold outside air into a plastic tube running along the greenhouse ceiling, allowing the air to warm up slightly before it descends to the plant zone.

Horizontal Air Flow (HAF) Fans: Fans are positioned to move air horizontally along the length of the greenhouse, blending cool air from vents with the existing warm air to prevent cold spots.

13.

Training	Pruning
Directing **shape and structure** (framework).	Managing **physiological functions** and fruit production.
Guiding growth via tying, trellising, or staking.	Selective removal of plant parts (leaves, stems, suckers).
Starts shortly after transplanting to establish a "skeleton".	Ongoing throughout the plant's life to maintain health and yield.
Improves light penetration and air circulation.	Redirects energy (sap flow) from foliage to flowers and fruit.

14. Microprocessors and computers act as the "brain" of a modern greenhouse, transforming it into a self-regulating ecosystem by continuously monitoring and adjusting environmental variables.

Sensors: Gather analog signals .

Analog-to-Digital Converter (ADC): Converts sensor readings into a digital format the microprocessor can process.

Microprocessor/Microcomputer: Executes programmed logic to decide whether to activate or deactivate equipment.

Actuators & Relays: Physical components (fans, heaters, sprinklers) that carry out the processor's commands.

PART C

15. 1. Organic Substrates (Bio-based)

- **Cocopeat (Coir): Derived from coconut husks; popular for its high water retention and durability.**
- **Peat Moss: Decomposed plant matter with high moisture holding and acidity; excellent for seedlings.**

- **Rice Hulls & Bark: Used primarily in mixes to improve drainage and reduce weight.**

2. Inorganic Substrates (Mineral-based)

- **Rockwool: Spun volcanic rock; allows for high-precision nutrient control and excellent oxygenation.**
- **Perlite: Expanded volcanic glass; used to increase aeration and prevent soil compaction.**
- **Vermiculite: A mica-like mineral that holds more water than perlite, ideal for moisture-loving plants.**
- **Expanded Clay: Porous pebbles used in hydroponics for their stability and high air content.**

16. Effective greenhouse problem management focuses on proactive sanitation, environmental control, and Integrated Pest Management (IPM) to prevent, detect, and treat issues like pests, diseases, and equipment failures. Key strategies include cleaning, scouting, and using biological controls, supported by maintaining proper humidity and temperature to reduce pathogen growth.

Use treated root substrates.